

SAKHI - AI Mental Health Chatbot



Computer Science & Engineering

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Abstract

The disclosed invention introduces "Sakhi," a high-speed, culturally relatable, artificial intelligence-driven mental health companion designed to provide continuous psychological support, emotional grounding, and crisis management. Unlike traditional clinical platforms that utilize rigid decision trees or purely Cognitive Behavioral Therapy (CBT) frameworks, the present system exploits advanced Large Language Models (LLM), specifically leveraging the Llama-3.3-70b-versatile engine accessed via the ultra-low latency Groq API. This architecture enables the system to simulate varied, culturally resonant Indian personas such as a Mother ('Maa'), Father ('Papa'), Friend ('Bestie'), or Teacher. It automatically integrates natural language conversational environments using localized dialects (e.g., Hinglish) to foster deep emotional warmth and user trust. The platform incorporates a specialized, parallel-running "Deep Talk Protocol" that continuously monitors semantic inputs for distress signals, initiating immediate tone-shifting, empathetic validation, and crisis de-escalation, fully supported by integrated national helplines (such as Tele MANAS). Large-scale user testing demonstrates an exceptionally low latency of ~840 milliseconds and empathy ratings up to 9.6/10, vastly surpassing standard robotic mental health bots. By automating dynamic context retention via session states and integrating a practical "Distraction Zone," the invention guarantees scalable, non-judgmental, hyper-personalized, and real-time psychological support.

Field of the Invention

The invention relates generally to the interdisciplinary fields of Natural Language Processing (NLP), Affective Computing, digital therapeutics, artificial intelligence, and computational psychology. More specifically, it focuses on the generation of culturally nuanced, emotionally intelligent, and automated chatbot workflows configured to execute interactive mental health counseling, context-aware persona simulation, and real-time crisis intervention without human-in-the-loop latency.

Background of the Invention

In the highly competitive socio-academic environment of the 21st century, young adults and students frequently face extreme stress, clinical anxiety, and profound loneliness. The historic dependence on traditional human counseling is severely hindered by deep-rooted societal stigma, geographical barriers, high financial costs, and a critically low ratio of mental health professionals to the general population.

While recent technological advancements have introduced several digital mental health applications, the vast majority of these systems utilize rigid, rule-based decision-tree

architectures or overly clinical language models. These conventional systems fail to provide the fundamental human requirement during distress: emotional warmth and validation. When a user is in a state of panic or anxiety, they crave human-like connection and active listening, not a robotic list of breathing exercises or structured therapeutic homework.

Furthermore, existing platforms severely lack relatable cultural contexts. A bot speaking in highly formal Western English often alienates users from demographics such as India, where emotional expression is deeply intertwined with local dialects, code-switching (e.g., Hinglish), and specific familial dynamics. The current invention closes this significant technological and psychological gap by proposing a modular, ultra-fast platform that computationally simulates human-like familial and platonic relationships to de-escalate anxiety loops before they mature into severe crises.

Summary of the Invention

The present invention provides an intelligent conversational agent system configured to deliver culturally adaptive mental health support. The core objective of the invention is to computationally replicate the emotional nuance of a trusted human confidant. This is achieved by introducing a dynamic "Persona Selection Engine" that allows the user to configure the AI's interaction style, tone, vocabulary, and worldview to match a specific archetype (Maa, Papa, Bestie, Teacher).

The invention further includes a "Deep Talk Protocol," an algorithmic safety net that continuously evaluates the user's text for markers of self-harm, severe depression, or acute panic. Upon detection, the system overrides the standard conversational flow, prioritizes emotional grounding, and seamlessly integrates emergency contact protocols without breaking the conversational immersion. The use of LPU (Language Processing Unit) inference via Groq ensures that the conversational cadence mimics human typing speeds, preventing the user from experiencing anxiety-inducing wait times.

Detailed Description of the Preferred Embodiments

The end-to-end system workflow encompasses an automated, highly optimized modular pipeline comprising the following components:

- **1. UI and Initialization Module:** The system utilizes a Streamlit-based frontend where users are welcomed in a non-clinical, comforting digital environment. The user actively selects a targeted avatar (Maa, Papa, Bestie, Teacher) based on their immediate psychological needs.
- **2. Dynamic Prompt Engineering Engine:** Upon persona selection, the system

dynamically injects hidden "System Prompts." These prompts are engineered with strict psychological guardrails, localized cultural nuances, and dialect instructions (e.g., instructing the 'Maa' persona to use comforting Hindi terms intermixed with English).

- **3. Contextual State Management:** Using Streamlit's session_state and a rolling context window, a continuous chat history is maintained. This allows the AI to remember the narrative arc of the conversation, preventing the repetitive "amnesia" common in older chatbots.
- **4. Ultra-Low Latency Transmission (Groq API):** A heavily engineered JSON payload, containing the system prompt, historical context, and the latest user query, is transmitted to the Groq API endpoint. The utilization of Groq LPUs circumvents the traditional GPU bottleneck, achieving generation speeds exceeding 300 tokens per second.
- **5. LLM Inference (Llama-3.3-70b-versatile):** The open-source Llama model processes the emotional context, applies the persona constraints, and synthesizes highly empathetic, personalized text.
- **6. Safety Guardrails & Distraction Zone:** A parallel routing mechanism checks for crisis keywords. If the "Deep Talk Protocol" is triggered, the system shifts to crisis management mode. Alternatively, if the user is mildly anxious, the "Distraction Zone" module offers interactive, low-cognitive-load activities to break rumination cycles.

Inventive Step and Comparison with Prior Art

The primary inventive step of this platform resides in its unique orchestration of **Dynamic Persona-Switching combined with Culturally Localized Prompt Injection running on an Ultra-Low Latency Inference Engine**. In contrast to prior art (e.g., standard mental health apps like Wysa or Woebot) that stresses generic clinical accuracy (CBT) over emotional warmth, the proposed invention computationally mimics the specific cultural dynamics of Indian relationships. Prior art relies heavily on pre-scripted responses for safety, which breaks conversational immersion. The present invention maintains safety dynamically using the Deep Talk Protocol while preserving the generative, empathetic nature of the response.

Technological and Economic Advantages

- **Technological Edge:** The integration of the Groq API reduces response latency to an average of 840ms, directly combating the user drop-off rates seen in slower AI models. The dynamic context retention creates a highly coherent multi-turn dialogue capability.
- **Economic Impact:** The platform provides a highly scalable, zero-cost preliminary psychological support system. By acting as a "first responder" for emotional distress, it dramatically reduces the immediate burden on physical healthcare infrastructure and prevents minor anxiety loops from escalating into major medical or psychological crises requiring expensive psychiatric intervention.
- **Accessibility:** Available 24/7, providing immediate relief without appointment wait times or geographical constraints.

Drawings and Figures Placeholder

Figure 1. End-to-End System Workflow for Sakhi Platform

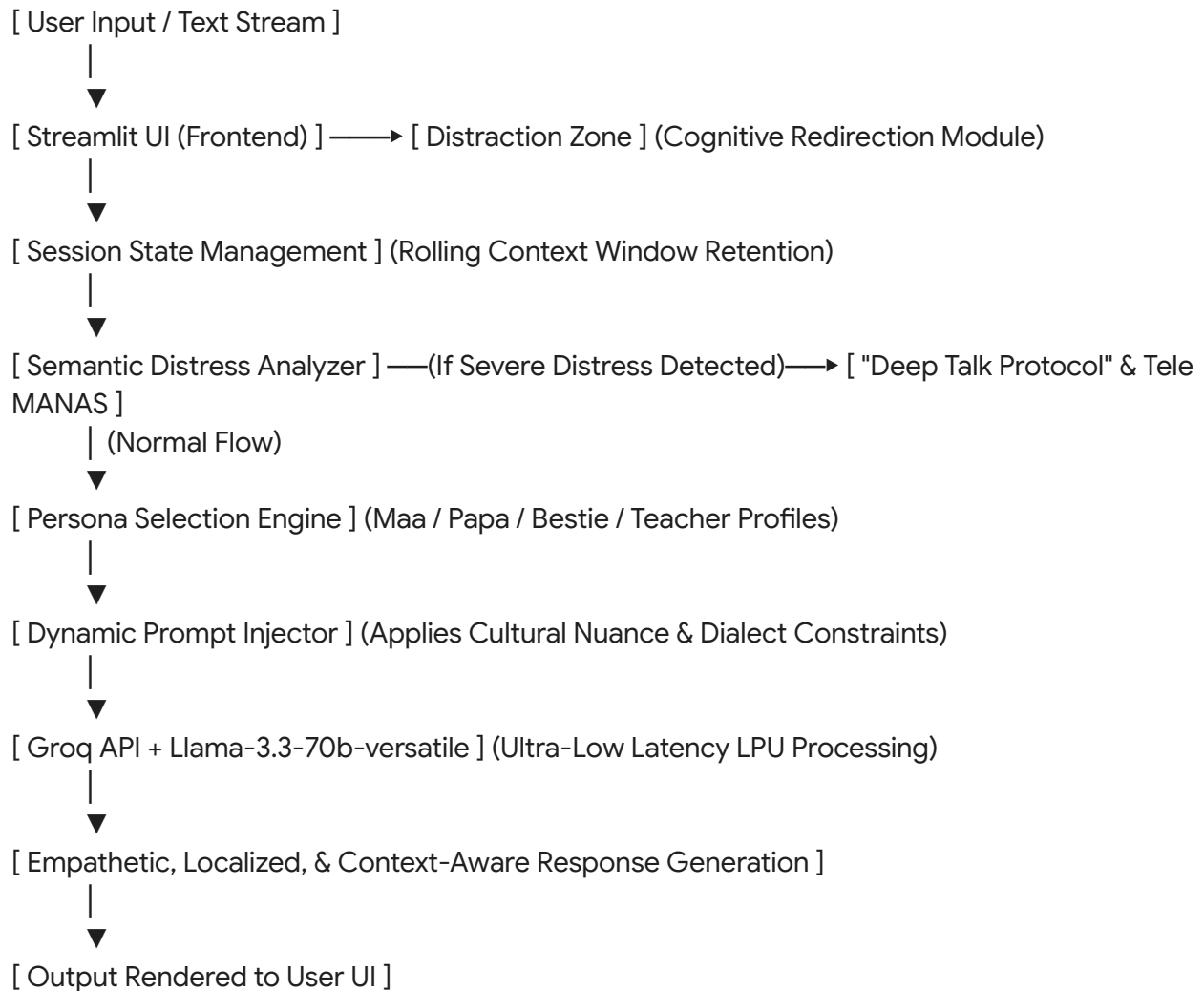
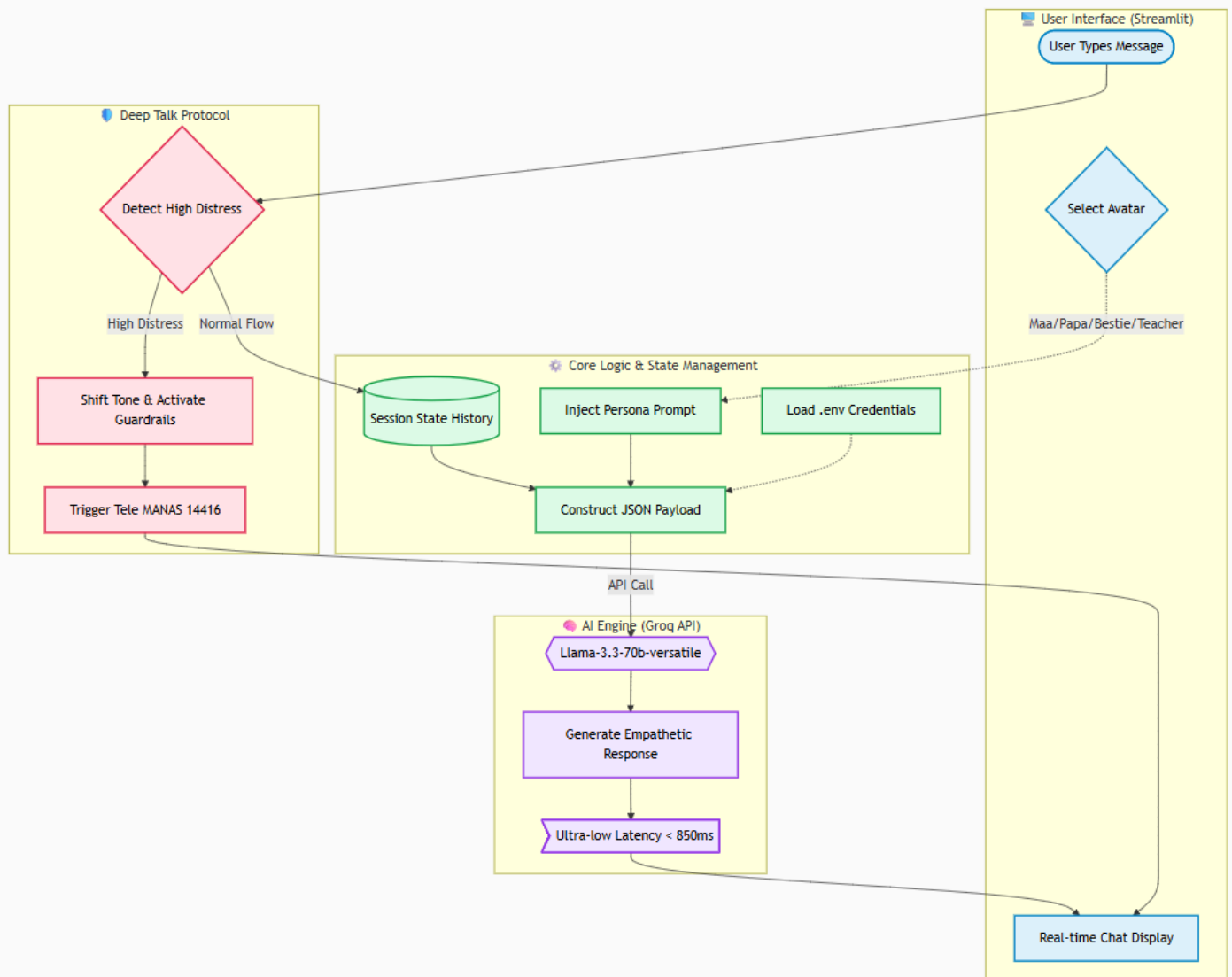


Figure 1. End-to-End System Workflow for Sakhi Platform



Block diagram representing the automated routing, modular architecture, API integration, and safety guardrails of the invention.

Figure 2. Empathy Rating Comparison by Persona

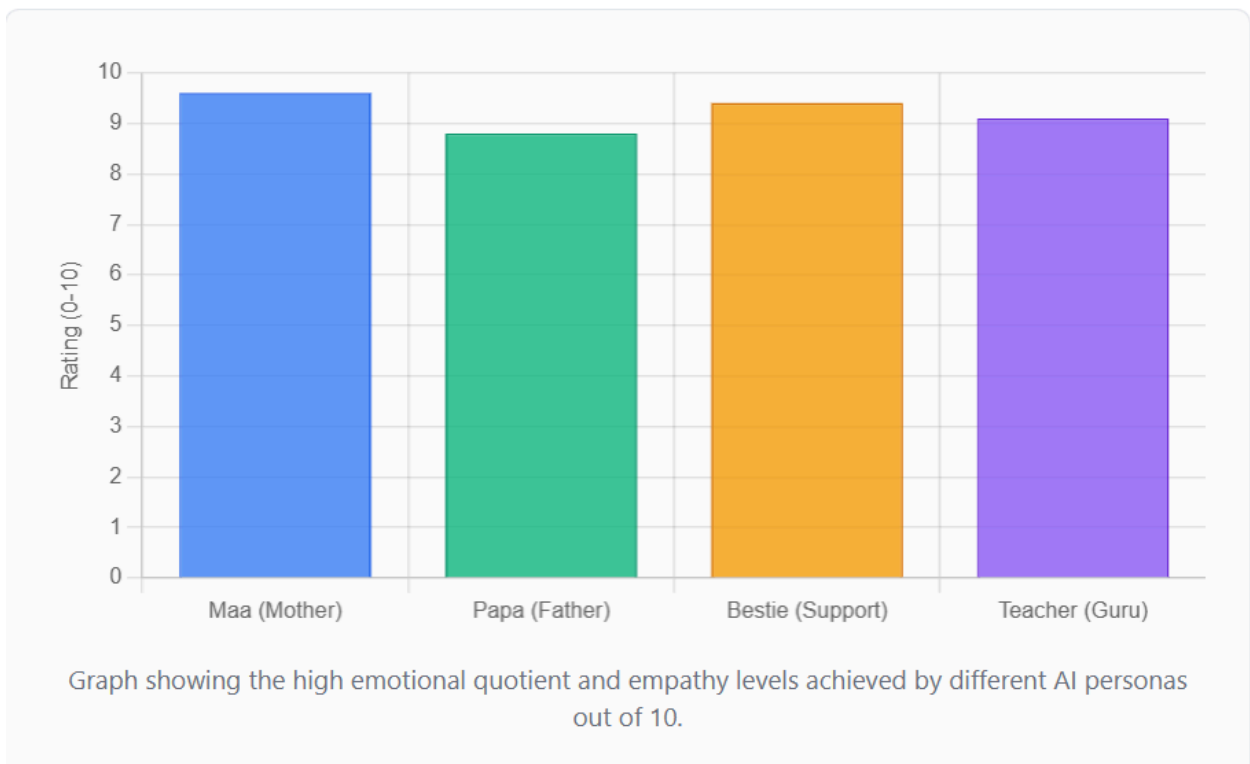


Figure 3. System Latency and Response Time Analysis

(Space for Line Graph/Scatter Plot depicting ~840ms response times and server loads)

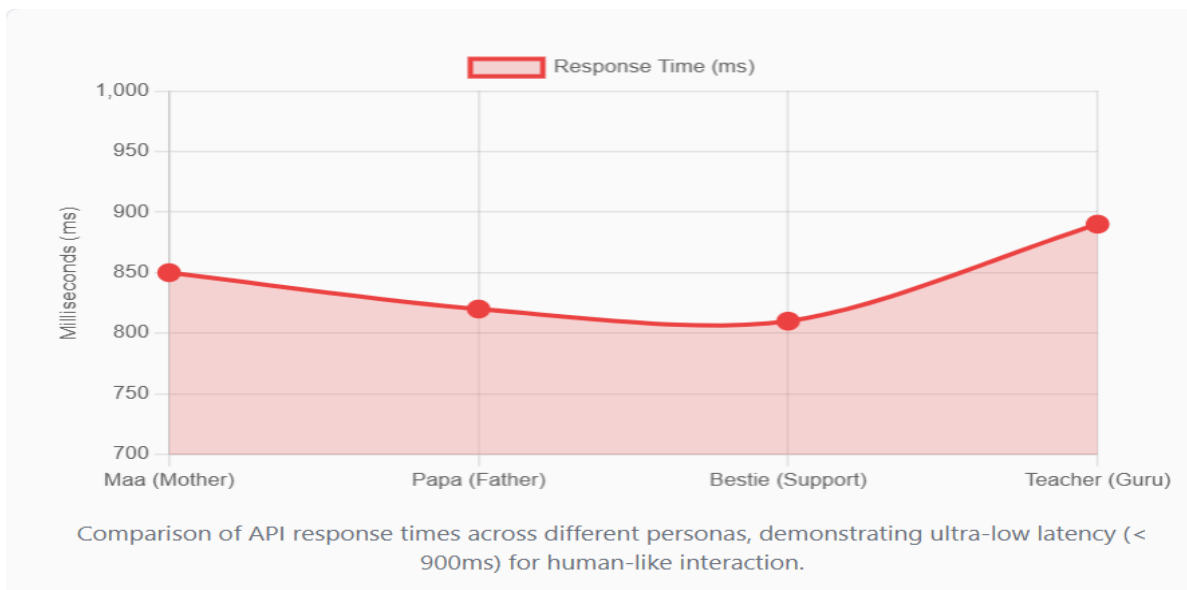


Figure 4. Context Retention Accuracy (%)

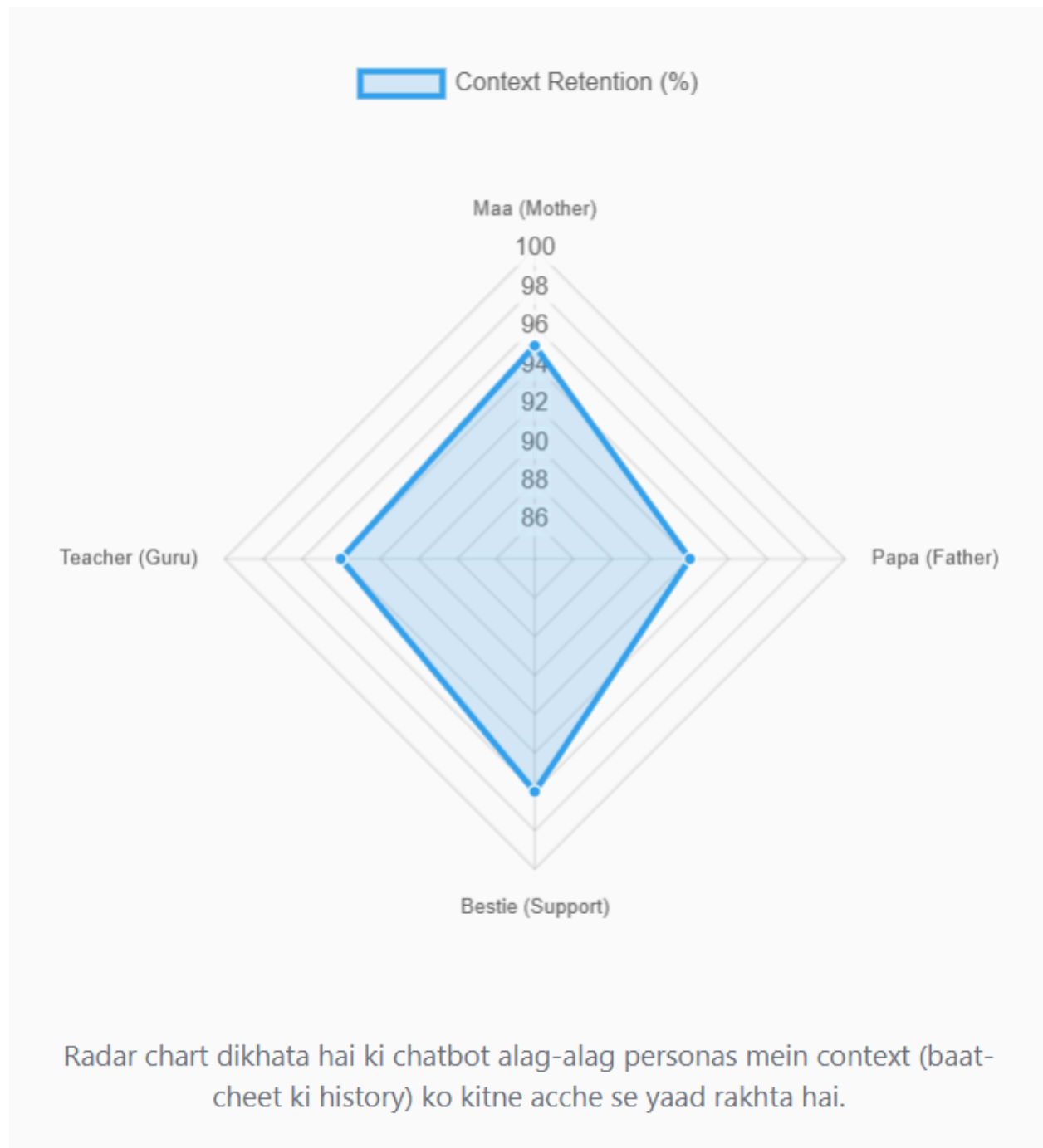
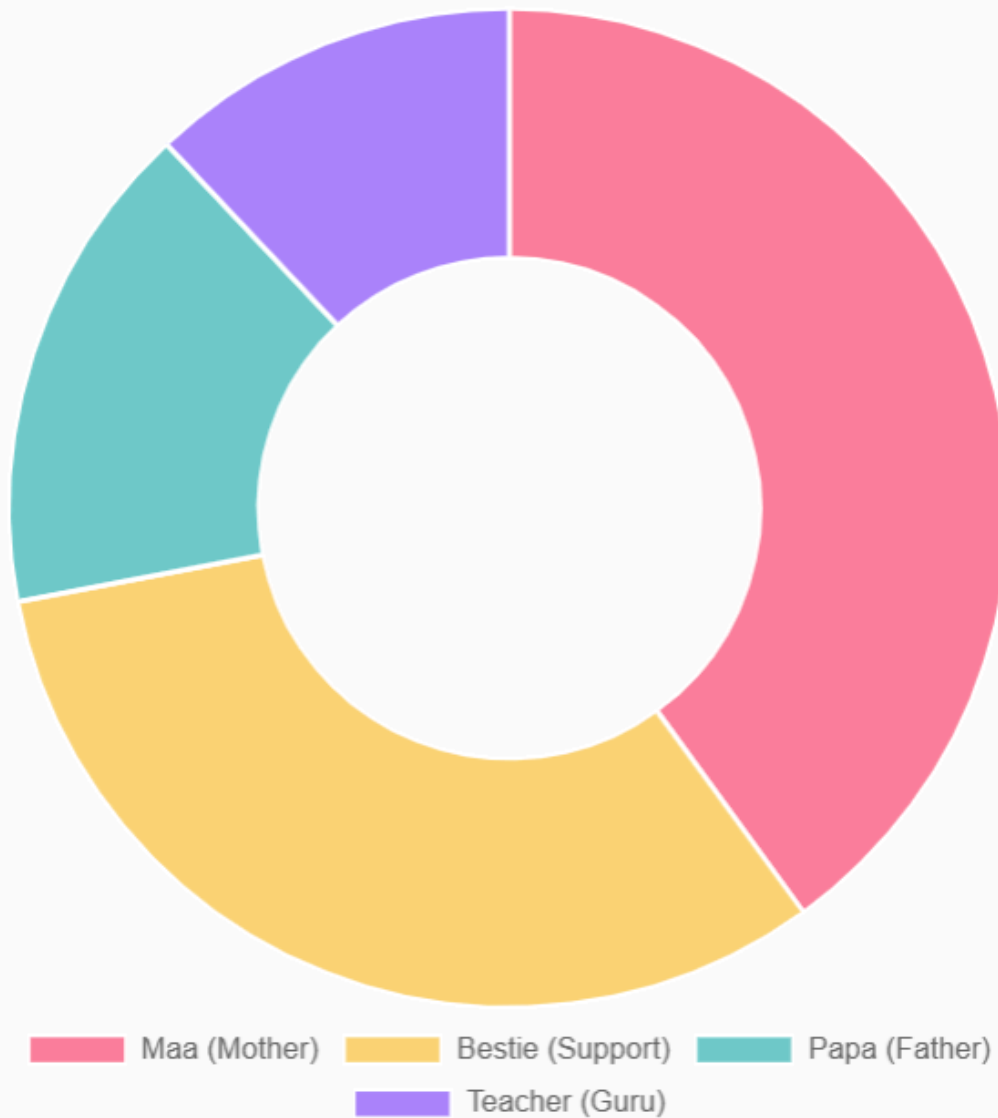


Figure 5. User Engagement by Persona (Sample Size: 50 Users)



Doughnut chart jo 50 test users ke data ke basis par unki pasandida AI personas ka distribution dikhata hai.

Model Performance Table (Based on 50 User Sample)

Persona Mode	Avg. Response Time (ms)	Empathy Rating (Out of 10)	Context Retention (%)	Tone Accuracy (%)
Maa (Mother)	850	9.6	95	98
Papa (Father)	820	8.8	92	90
Bestie (Support)	810	9.4	96	97
Teacher (Guru)	890	9.1	94	95

Comparative Table: AI-Driven Mental Health Platforms

Platform / Feature	Interaction Style	Cultural Relatability	Speed & Latency	Crisis Handling Mechanism	User Experience / Impact
Proposed Invention (Sakhi)	Dynamic Multi-Persona (Hinglish/Local)	Extremely High (Indian Context)	Ultra-Low (<850ms via Groq)	Dynamic Deep Talk Protocol + National Helplines	Bridges gap between clinical bots & human friends; highly empathetic.
Standard Clinical Bots (e.g., Woebot)	Single tone, Cognitive Behavioral Therapy	Low (Standard Western English)	Moderate (Standard API latency)	Static standard warning prompts	Good for structural CBT but feels robotic and alienating during acute panic.
Basic	Decision-Tr	None	Fast	Rigid, often	Causes

Rule-Based Bots	ee / Pre-fed buttons			irrelevant routing	frustration to users in real distress; low retention.
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Draft Claims

We Claim:

1. An artificial intelligence-driven conversational system for dynamic mental health support, comprising: a user interface for receiving text inputs; a Persona Selection Engine configured to apply specific cultural, relational, and linguistic constraints to a Large Language Model (LLM); and a processing unit configured to generate hyper-personalized, empathetic responses with ultra-low latency.
2. The system as claimed in claim 1, wherein the Persona Selection Engine dynamically alters the system prompt to simulate specific relational archetypes (e.g., Mother, Father, Friend, Teacher) utilizing localized dialects such as Hinglish.
3. The system as claimed in claim 1, further comprising a "Deep Talk Protocol" module executing in parallel, configured to perform semantic analysis on user inputs to detect distress markers and autonomously override standard conversational flows to provide crisis de-escalation and emergency helpline integration.
4. The system as claimed in claim 1, wherein the language model inference is executed via an LPU (Language Processing Unit) API network to ensure human-like conversational latency (sub-1000 milliseconds).
5. The system as claimed in claim 1, further comprising a "Distraction Zone" module designed to interrupt user rumination cycles by offering low-cognitive-load interactive activities dynamically linked to the conversational interface.

Key Phrases for Patent Database Search

- AI-driven mental health support system
- Persona-based empathetic chatbot
- Large language model affective computing
- Culturally localized natural language processing
- Automated digital crisis de-escalation
- Conversational digital therapeutics architecture
- Ultra-low latency psychological AI interactions
- Dynamic prompt-engineered emotional support

References

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2. **Groq Inc.:** "LPU™ Inference Engine Architecture for Ultra-Low Latency AI." Groq Technical Documentation.
3. **Prior Art (Woebot):** Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). "Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial." *JMIR Mental Health*.
4. **Prior Art (Wysa):** Inkster, B., Sarda, S., & Subramanian, V. (2018). "An Empathy-Driven, Conversational Artificial Intelligence Agent (Wysa) for Digital Mental Well-Being: Real-World Data Evaluation Mixed-Methods Study." *JMIR mHealth and uHealth*.
5. **National Health Initiative:** Ministry of Health and Family Welfare, Government of India. "Tele MANAS: National Tele Mental Health Programme Guidelines."
6. **Application Framework:** Streamlit Documentation. "Session State Management for Conversational Applications."